

Midterm

GRA4153 Advanced Statistics

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Question 1

Let X be a random variable and $\mu := \mathbb{E}[X]$; we are interested in bounds on the function

$$\alpha(t) := P(|X - \mu| > t),$$

for $t > 0$

- (i) In class we used Markov's inequality to derive a bound on $\alpha(t)$. Derive such a result, under the assumption that X has a finite p -th moment (for $p \geq 1$), i.e. $\mathbb{E}|X|^p < \infty$.

By Markov's inequality applied to $Y = |X - \mu|^p$, [Lee25a, THEOREM 2.3]

$$P(|X - \mu| > t) = P(|X - \mu|^p > t^p) \leq \frac{\mathbb{E}|X - \mu|^p}{t^p}.$$

□

- (ii) We shall provide a much better bound for the case where $X \sim \mathcal{N}(\mu, \sigma^2)$. In particular we will show that

$$\alpha(t) \leq \frac{\sigma}{t} \frac{2}{\sqrt{2\pi}} \exp(-t^2/(2\sigma^2)) \quad (1)$$

- (a) We consider first the case where $X \sim \mathcal{N}(0, 1)$, Express α as an integral [Lee25a, EXAMPLE 2.6].

$$\alpha(t) = P(|X| > t) = P(X > t) + P(X < -t) = 2P(X > t) = 2 \int_t^\infty \frac{1}{\sqrt{2\pi}} \exp(-x^2/2) dx$$

- (b) Use the change of variables $x = t + y$ to simplify the integral and attain (1) for the standard normal case.

We use the substitution $x = t + y$ (so $dx = dy$) to get

$$\begin{aligned} \alpha(t) &= \frac{2}{\sqrt{2\pi}} \int_0^\infty \exp(-(t+y)^2/2) dy = \\ &= \frac{2}{\sqrt{2\pi}} \exp(-t^2/2) \int_0^\infty \exp(-ty - y^2/2) dy \leq \frac{2}{\sqrt{2\pi}} \exp(-t^2/2) \int_0^\infty \exp(-ty) dy = \\ &= \frac{2}{\sqrt{2\pi}} \exp(-t^2/2) \cdot \frac{1}{t}, \quad t > 0 \end{aligned}$$

Thus, for $X \sim \mathcal{N}(0, 1)$,

$$\alpha(t) \leq \frac{2}{t\sqrt{2\pi}} \exp(-t^2/2)$$

(c) For the general case $X \sim \mathcal{N}(\mu, \sigma^2)$, let $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$. Then

$$\alpha(t) = P(|X - \mu| > t) = P(|Z| > t/\sigma) \leq \frac{2}{(t/\sigma)\sqrt{2\pi}} \exp\left(-\frac{(t/\sigma)^2}{2}\right) = \frac{\sigma}{t} \frac{2}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

Comparison with part (i): The bound in (i) decays polynomially as t^{-p} , while the Gaussian bound decays exponentially in t^2 , which is dramatically sharper for large t .

(iii) Now we'll derive a slightly looser exponential bound for the normal distribution using an alternative approach which is more easily generalizable. Specifically, this approach is applicable whenever the MGF exists. In the Gaussian case we get:

$$\alpha(t) \leq 2 \exp(-t^2/(2\sigma^2)) \quad (2)$$

(a) We consider first the case where $X \sim \mathcal{N}(0, 1)$. We will use Markov's inequality, but in a clever way: argue that, for $\lambda > 0$,

$$P(X > t) = P(\exp(\lambda X) > \exp(\lambda t))$$

Because the exponential is strictly increasing when $\lambda > 0$, the events are identical. For any real numbers $x, t, \lambda > 0$, $x > t \iff \exp(\lambda x) > \exp(\lambda t)$. Therefore, for any random variable X and $t \in \mathbb{R}$, $P(X > t) = P(\exp(\lambda X) > \exp(\lambda t))$

(b) Use Markov's inequality to provide a bound for the preceding display.

We apply Markov to the nonnegative r.v. $Y = \exp(\lambda X)$:

$$P(X > t) = P(\exp(\lambda X) > \exp(\lambda t)) \leq \frac{\mathbb{E}[\exp(\lambda X)]}{\exp(\lambda t)} = M_X(\lambda) \exp(-\lambda t).$$

The MGF of $\mathcal{N}(0, 1)$ [Lee25b, Appendix A, PROPOSITION 2] is

$$M_X(\lambda) = \exp(\lambda^2/2).$$

Hence,

$$P(X > t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{2}\right)$$

(c) Choose the best possible $\lambda > 0$ and use this to provide a bound for $\alpha(t)$

Here we need to optimize over $\lambda > 0$, i.e. we need to derive $f(\lambda) = -\lambda t + \frac{\lambda^2}{2} \rightarrow f'(\lambda) = -t + \lambda = 0$; the exponent $-\lambda t + \lambda^2/2$ is minimized at $\lambda = t$, giving

$$P(X > t) \leq \exp(-t^2/2) \Rightarrow \alpha(t) = 2P(X > t) \leq 2 \exp(-t^2/2)$$

(d) Use the bound for the case $\mathcal{N}(0, 1)$ to derive a bound for the case $\mathcal{N}(\mu, \sigma^2)$

Comment on how this approach could be used to bound $\alpha(t)$ for random variables other than $X \sim \mathcal{N}(\mu, \sigma^2)$.

We have the standard normal from c:

$$\alpha(t) = 2P(X > t) \leq 2 \exp(-t^2/2)$$

Then we can plug in the standardized variable:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Hence, for any $t > 0$,

$$\alpha(t) = P(|X - \mu| > t) = P\left(\left|\frac{X - \mu}{\sigma}\right| > \frac{t}{\sigma}\right) = P(|Z| > t/\sigma) \leq 2 \exp\left(-\frac{1}{2}(t/\sigma)^2\right) = 2 \exp\left(-\frac{t^2}{2\sigma^2}\right)$$

Equivalently, using the Gaussian MGF $M_X(\lambda) = \exp(\lambda\mu + \frac{1}{2}\lambda^2\sigma^2)$ and the shift-scaling property $M_{aX+b}(t) = \exp(bt)M_X(at)$ [Lee25b, Appendix A, LEMMA 2 and PROPOSITION 2], the same bound follows.

Generalization beyond Gaussian: If X has an MGF in a neighborhood of 0, Chernoffs method gives, for any $\lambda > 0$,

$$P(X - \mu \geq t) \leq \exp(\psi(\lambda) - \lambda t), \quad \psi(\lambda) := \log \mathbb{E}[\exp(\lambda(X - \mu))],$$

and the tightest bound is obtained by minimizing the RHS over $\lambda > 0$. A two-sided bound follows by symmetry or by applying the same argument to $-(X - \mu)$. This is the standard Chernoff bound technique and applies widely to sub-Gaussian and other light-tailed distributions [Ver18, Chapter 2, sec. 2.3].

Question 2

We test $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \in \Theta_1$, with a continuous statistic $T(X)$ where larger T favors H_1 .

- (i) Show that $p(x) = P_{\theta_0}(T(X) \geq T(x))$ is a valid p-value.

Let F_0 be the CDF of $T(X)$ under H_0 . For the observed x ,

$$p(x) = P_{\theta_0}(T(X) \geq T(x)) = 1 - F_0(T(x)).$$

Thus, $p(x) = 1 - F_0(T(x))$. Let $U := F_0(T(X))$. By the probability integral transform [Lee25b, Appendix B, THEOREM 1], $U \sim U(0, 1)$, hence $p(X) = 1 - U \sim U(0, 1)$. Therefore $P_{\theta_0}(p(X) \leq \alpha) = \alpha \quad \forall \alpha \in [0, 1]$ if F_0 is continuous; without continuity one has $P_{\theta_0}(p(X) \leq \alpha) \leq \alpha$, so p is a valid p-value.

- (ii) Show that $\phi(X) = \mathbb{1}\{p(X) \leq \alpha\}$ is a level α test for H_0 against H_1 .

The definition of the level- α test [Lee25a, 3.3.4: Hypothesis testing] states that a (non-randomized) test is a function

$$\phi : \mathcal{X} \rightarrow \{0, 1\}, \quad \phi(x) = 1 \text{ means reject } H_0.$$

The size of a test is the greatest probability of rejecting H_0 when H_0 is true:

$$\sup_{\theta \in \Theta_0} \mathbb{E}[\phi(X)] = \sup_{\theta \in \Theta_0} P_{\theta}\{\phi(X) = 1\}.$$

A test is said to be level α if its size does not exceed α . For a simple null hypothesis $H_0 : \theta = \theta_0$, this reduces to:

$$P_{\theta_0}\{\phi(X) = 1\} \leq \alpha.$$

Under H_0 ,

$$\mathbb{E}_{\theta_0}[\phi(X)] = P_{\theta_0}(p(X) \leq \alpha) = \alpha \quad (\text{since we have continuity in (i)}),$$

so ϕ is level α .

- (iii) Consider the special case where $X \sim \mathcal{N}(\theta, 1)$ and we want to test $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$. Provide an explicit formula for a valid p-value.

As emphasized in the notes [Lee25a, p. 81], we choose a statistic that is large under H_1 and small under H_0 ; by symmetry of $\mathcal{N}(0, 1)$ and the two-sided alternative, $T(X) = |X|$ is the natural choice. Under H_0 , $Z := X \sim \mathcal{N}(0, 1)$. By the definition $p(x) = P_0(T(Z) \geq T(x))$, using $T(X) = |X|$ (and equivalently, $T(X) = X^2$):

$$p(x) = P(|Z| \geq |x|) = 2(1 - \Phi(|x|)) = 2\Phi(-|x|),$$

where Φ is the standard normal CDF.

Equivalently, using $T(X) = X^2$ (so $T(Z) \sim \chi^2$ with 1 degree of freedom):

$$p(x) = P(\chi_1^2 \geq x^2).$$

Because the null distribution is continuous, this p -value is exact, and hence valid.

- (iv) Suppose that for each $\alpha \in [0, 1]$ a critical value c_α can be chosen s.t. $\psi_\alpha(X) = \mathbf{1}\{T(X) \geq c_\alpha\}$ is a size α test of H_0 against H_1 . Use this family of tests to show that the p -value defined in (i) is the smallest α level at which we can reject H_0 , having observed the statistic $T(x)$.

Assume for each $\alpha \in [0, 1]$ $\exists c_\alpha$ with $\psi_\alpha(X) = \mathbf{1}\{T(X) \geq c_\alpha\}$ a size- α test. Under continuity, size α means

$$P_{\theta_0}(T(X) \geq c_\alpha) = \alpha \quad \Rightarrow \quad c_\alpha = F_0^{-1}(1 - \alpha).$$

Then, for observed $T(x)$,

$$\psi_\alpha(x) = 1 \iff T(x) \geq c_\alpha \iff F_0(T(x)) \geq 1 - \alpha \iff \alpha \geq 1 - F_0(T(x)) = p(x).$$

Thus the smallest α for which ψ_α rejects is exactly $p(x)$

Question 3

We observe independent Bernoulli samples $X_1, \dots, X_{n_1}$ with mean p_1 and $Y_1, \dots, Y_{n_2}$ with mean p_2 , and set

$$\hat{p}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} X_i, \quad \hat{p}_2 = \frac{1}{n_2} \sum_{i=1}^{n_2} Y_i, \quad \hat{p} = \frac{1}{n_1 + n_2} \left[\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i \right]$$

Let $s_i^2 := \frac{1}{n_i} \sum_{j=1}^{n_i} (W_{ij} - \hat{p}_i)^2$ for $W = X$ for $i = 1$ and $W = Y$ for $i = 2$. Define

$$\hat{Z}_1 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad \hat{Z}_2 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}}, \quad \hat{Z}_3 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

(i) Comment on the difference between the three tests. Ideally you should explain *why* these tests have different forms in the denominator.

- $\hat{Z}_1$ (pooled): Uses the pooled estimator of the common variance under H_0 , namely $p(1-p)(1/n_1 + 1/n_2)$ with p estimated by the pooled proportion $\hat{p}$. This enforces H_0 in the variance estimate and is the classical two-sample proportion z-test for hypothesis testing, namely because it uses the pooled MLE. The main difference between this estimator and the others mentioned, is that it is more robust under H_0 .
- $\hat{Z}_2$ (unpooled): Uses a plug-in variance that does not impose H_0 , so it is natural for confidence intervals on $p_1 - p_2$. As a test for H_0 , it can be slightly conservative or anti-conservative in finite samples, especially when p is near 0 or 1 or the groups are unbalanced. In other words, this estimator aligns with interval estimation for $p_1 - p_2$.
- $\hat{Z}_3$ (studentized by sample variance): Here $s_i^2 = \frac{1}{n_i} \sum_{j=1}^{n_i} (W_{ij} - \hat{p}_i)^2$, $W = X$ if $i = 1$, $W = Y$ if $i = 2$ is the sample second central moment with denominator n_i (not $n_i - 1$). For Bernoulli data, $s_i^2 = \hat{p}_i(1 - \hat{p}_i)$ exactly, because $W_{ij}^2 = W_{ij}$.
 $\frac{1}{n_i} \sum_{j=1}^{n_i} (W_{ij} - \hat{p}_i)^2 = \frac{1}{n_i} \sum W_{ij} - \hat{p}_i^2 = \hat{p}_i - \hat{p}_i^2 = \hat{p}_i(1 - \hat{p}_i)$, therefore $\hat{Z}_3 \equiv \hat{Z}_2$, which implies that they will overlap. If one had used the unbiased sample variance with denominator $n_i - 1$, the difference would vanish asymptotically.

(ii) Explore the performance of these three tests in a simulation

Here we need to fix our M (I'll do 5000 because the question suggests it). Furthermore, we need to simulate M independent datasets for chosen (p_1, p_2, n_1, n_2) , and then compute $\hat{Z}_1, \hat{Z}_2, \hat{Z}_3$ for each, and apply the two-sided rule with critical value $c = z_{1-\alpha/2}$, and estimate rejection probabilities by the fraction of rejections. Then we'll repeat over a grid of scenarios to examine size ($p_1 = p_2$) and power ($p_1 \neq p_2$), plus the effect of n_1, n_2 and extremeness of p_1, p_2 .

The python file can be found as a separate document (`task.three.py`), which outputs

p_1	p_2	n_1	n_2	reject Z_1	reject Z_2	reject Z_3
0.10	0.10	50	50	0.048	0.057	0.057
0.10	0.10	200	200	0.048	0.051	0.051
0.10	0.10	50	200	0.049	0.074	0.074
0.50	0.50	50	50	0.057	0.057	0.057
0.50	0.50	200	200	0.050	0.050	0.050
0.50	0.50	50	200	0.055	0.057	0.057
0.90	0.90	50	50	0.050	0.059	0.059
0.90	0.90	200	200	0.047	0.051	0.051
0.90	0.90	50	200	0.048	0.071	0.071
0.10	0.20	50	50	0.308	0.314	0.314
0.10	0.20	200	200	0.815	0.815	0.815
0.10	0.20	50	200	0.365	0.523	0.523
0.40	0.60	50	50	0.541	0.542	0.542
0.40	0.60	200	200	0.980	0.980	0.980
0.40	0.60	50	200	0.728	0.728	0.728
0.70	0.50	50	50	0.543	0.546	0.546
0.70	0.50	200	200	0.984	0.984	0.984
0.70	0.50	50	200	0.746	0.762	0.762

What does all this mean?

In the H_0 cases (nominal $\alpha = 0.05$):

- Balanced, moderate n : $p_1 = p_2 = 0.50, n_1 = n_2 = 200 \rightarrow \hat{Z}_1 = 0.05, \hat{Z}_2 = 0.05, \hat{Z}_3 = 0.05$. Excellent calibration for all.
- Balanced, smaller n : $p_1 = p_2 = 0.10$ or $0.90, n_1 = n_2 = 50 \rightarrow \hat{Z}_1 \approx 0.048 - 0.05, \hat{Z}_2 = \hat{Z}_3 \approx 0.057 - 0.059$. $\hat{Z}_2/\hat{Z}_3$ a tad liberal.
- Unbalanced, extreme p : $p_1 = p_2 = 0.10$ or $0.90, n_1 = 50, n_2 = 200 \rightarrow \hat{Z}_1 \approx 0.048 - 0.049$ (on target), $\hat{Z}_2 = \hat{Z}_3 \approx 0.071 - 0.074$ (noticeably liberal).

H_1 cases (power):

- Balanced: $p_1 = 0.40, p_2 = 0.60, n_1 = n_2 = 200 \rightarrow \text{power} \approx 0.98$ for all
- Smaller n : $p_1 = 0.10, p_2 = 0.20$ or $0.90, n_1 = n_2 = 50 \rightarrow \text{power} \approx 0.308 - 0.314$ (very similar).
- Unbalanced and more extreme: $p_1 = 0.10, p_2 = 0.20, n_1 = 50, n_2 = 200 \rightarrow \hat{Z}_1 = 0.365$ vs $\hat{Z}_2 = \hat{Z}_3 = 0.523$. This big gap mirrors the size inflation $\hat{Z}_2/\hat{Z}_3$ showed under H_0 in similar settings; a fair power comparison should account for that miscalibration.

$\hat{Z}_2$ & $\hat{Z}_3$ use groupwise plug-in variances. In small or imbalanced samples, and especially when proportions are extreme, these plug-ins can be very small (even zero with all-0 or all-1 groups), shrinking the denominator and making $|\hat{Z}|$ too large, so the test is liberal. $\hat{Z}_1$ pools counts, smoothing those extremes and stabilizing the denominator under H_0 , so its size stays closer to nominal.

Rejection increases with $|p_1 - p_2|$ and with both n_1 and n_2 . For fixed total $n_1 + n_2$, balancing the groups ($n_1 = n_2$) maximizes power by minimizing $\frac{1}{n_1} + \frac{1}{n_2}$. For fixed $|p_1 - p_2|$ and sample sizes, power is typically higher when the proportions are farther from 0.5 (since $p(1-p)$ is smaller, standard errors are smaller), though $\hat{Z}_2$ & $\hat{Z}_3$ also show more size distortion in these edge cases.

References

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